

Technology Stack

Date	27 June 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	Streamlit	Enables rapid development of an interactive Python-based UI without separate frontend code, ideal for data/AI-driven applications

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python	Native compatibility with all required AI/ML libraries (Whisper, Sentence-BERT, Librosa); single-language stack simplifies development
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	JSON file (reference_concepts.json)	Lightweight, sufficient for storing static reference concept explanations; avoids the overhead of a full database for a small, fixed dataset
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Docker + Hugging Face Spaces	Free CPU-based hosting purpose-built for ML/AI demo applications; Docker ensures consistent environment setup, avoiding "works on my machine" issues
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git + GitHub	Standard for source control and collaboration; enables tracking of code changes and team contributions over time

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	OpenAI Whisper, Sentence-Transformers (Sentence-BERT), Librosa, Google Gemini API	Whisper performs speech-to-text transcription; Sentence-BERT computes semantic similarity; Librosa extracts audio fluency features; Gemini API generates natural-language feedback from raw scores
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